

LFP-8: 8-channel LiFePO4 32140/33140 charger / discharger / IR tester (5 boards = 40 cells)

Single source of truth: `hardware/gen/lfp8_circuit.py`. All nets are connected by labels; power nets use power symbols.

power

Power input, 3.3 V LDO, 2.495 V reference, ISET DAC

File: lfp8_power.kicad_sch

safety

SAFE_EN chain: watchdog, rail UV/OV, board over-temperature, E-STOP

File: lfp8_safety.kicad_sch

mcu

ESP32-C3 module, USB, buttons, fan

File: lfp8_mcu.kicad_sch

adc

ADS1115 + CD4051 measurement multiplexers

File: lfp8_adc.kicad_sch

logic

74HC595 control word, level shifters, SR output enable

File: lfp8_logic.kicad_sch

ch1

Cell channel 1: charger, discharger, protection, measurement

File: lfp8_ch1.kicad_sch

ch2

Cell channel 2: charger, discharger, protection, measurement

File: lfp8_ch2.kicad_sch

ch3

Cell channel 3: charger, discharger, protection, measurement

File: lfp8_ch3.kicad_sch

ch4

Cell channel 4: charger, discharger, protection, measurement

File: lfp8_ch4.kicad_sch

ch5

Cell channel 5: charger, discharger, protection, measurement

File: lfp8_ch5.kicad_sch

ch6

Cell channel 6: charger, discharger, protection, measurement

File: lfp8_ch6.kicad_sch

ch7

Cell channel 7: charger, discharger, protection, measurement

File: lfp8_ch7.kicad_sch

ch8

Cell channel 8: charger, discharger, protection, measurement

File: lfp8_ch8.kicad_sch

mech

Mounting holes

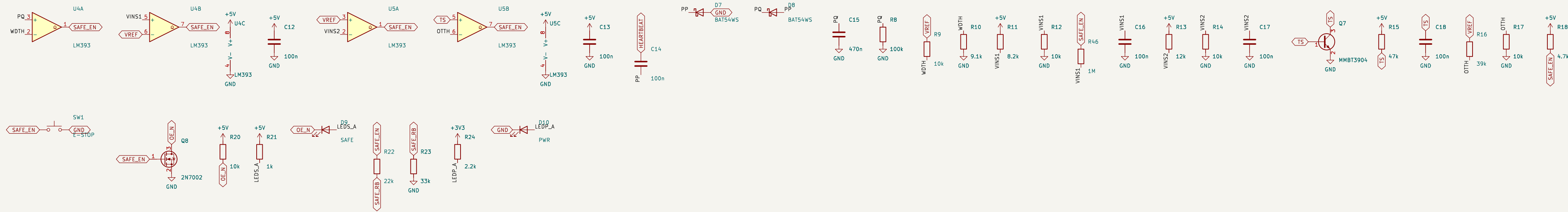
File: lfp8_mech.kicad_sch

Title: LFP-8 8-channel LFP cell tester – power		
Size: A2	Date: 2026-10-01	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/15

Size: A2	Date: 2026-10-01	Rev: 1.0
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SAFE_EN chain: watchdog, rail UV/OV, board over-temperature, E-STOP

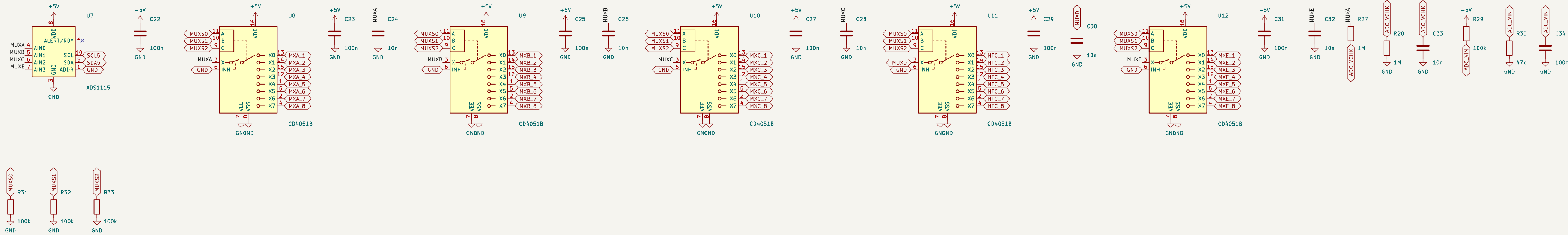


Generated by hardware/gen/make_sch.py from the netlist DSL – do not edit by hand
 ESP32–C3 module, USB, buttons, fan
github.com/maxakash/lfppcb
 Sheet: /mcu/
 File: lfp8_mcu.kicad_sch

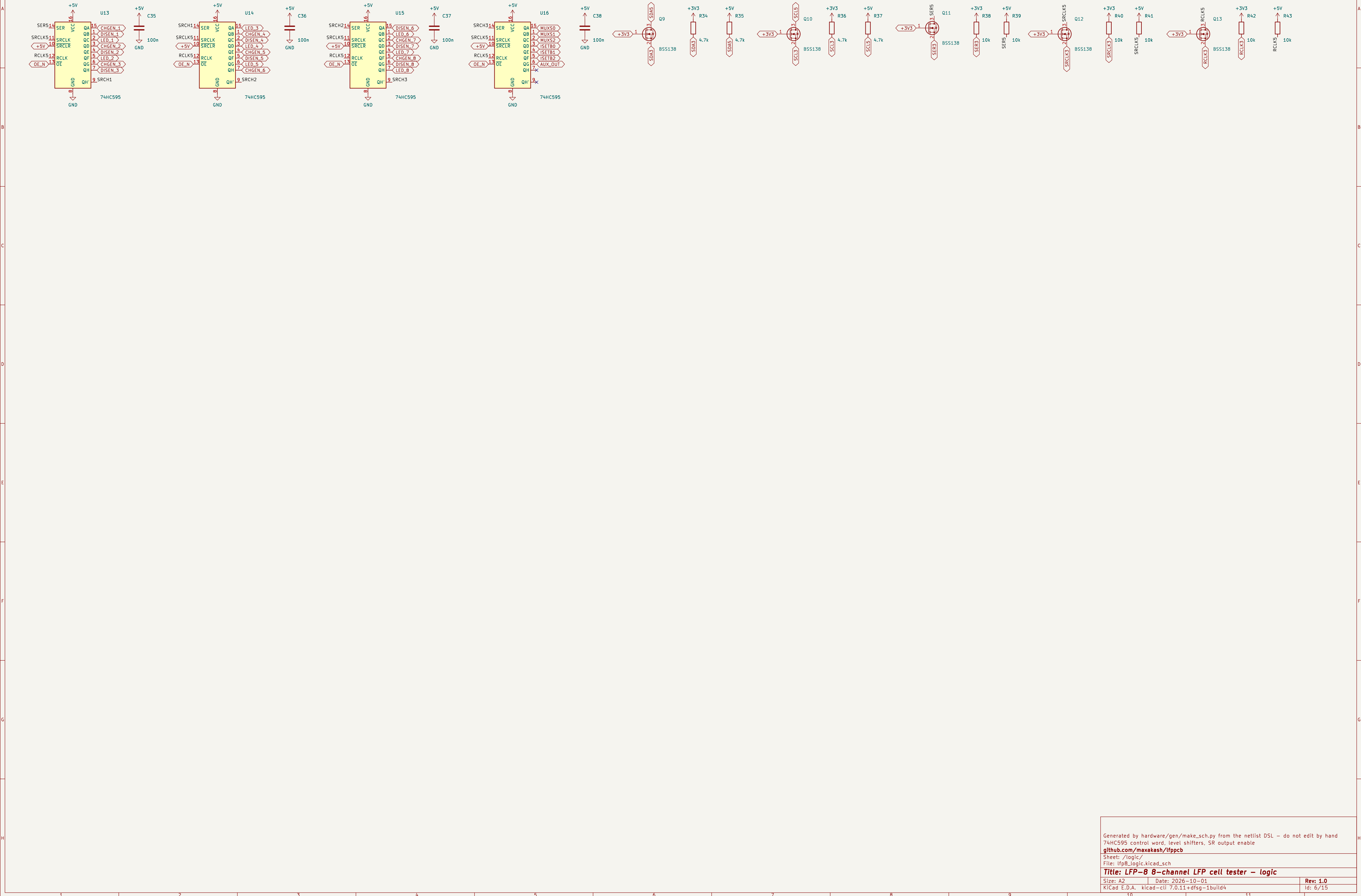
Title: LFP-8 8-channel LFP cell tester – mcu

Size: A2	Date: 2026-10-01	Rev: 1.0
KiCad EDA	kicad-ctl 7.0.11+dfsg-1build4	Id: 4/15

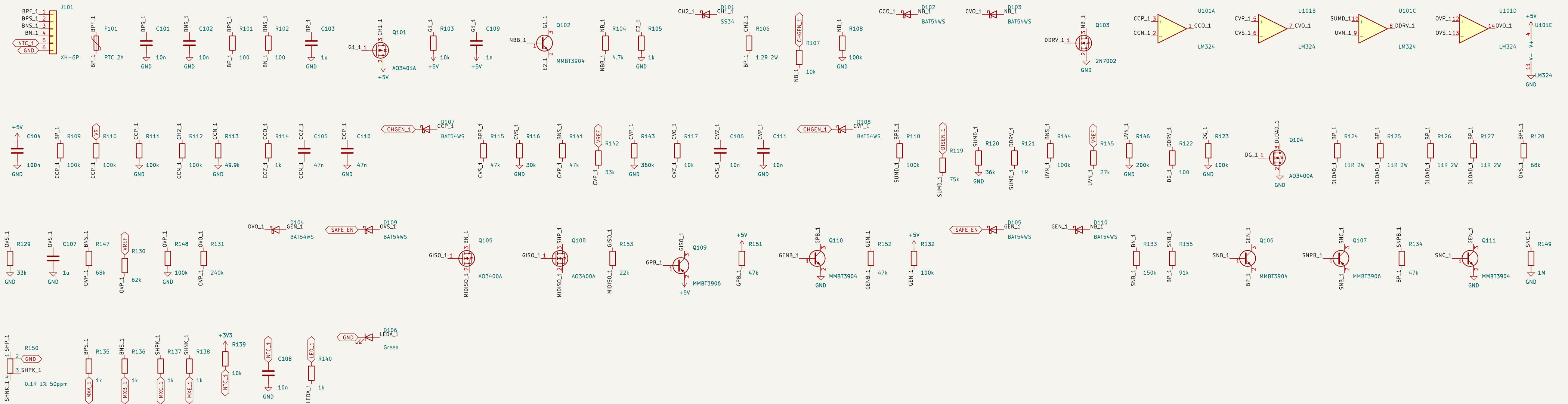
ADS1115 + CD4051 measurement multiplexers



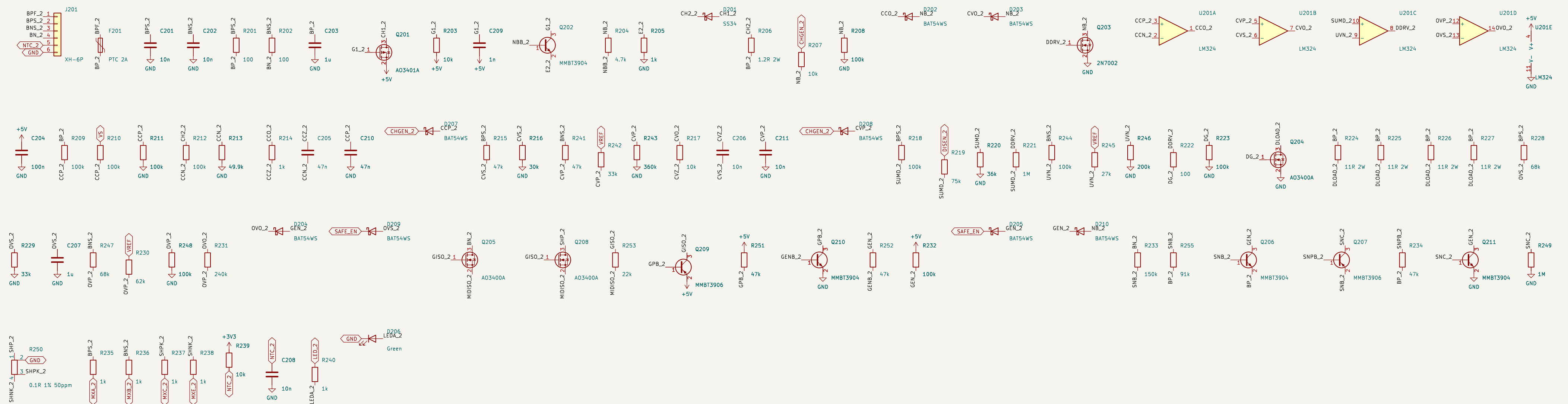
74HC595 control word, level shifters, SR output enable



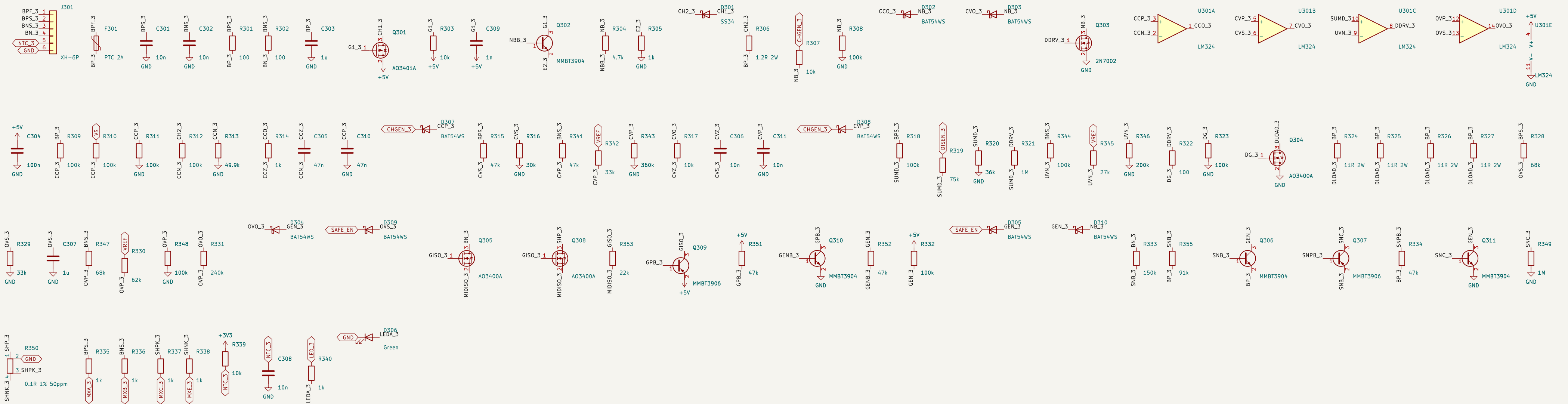
Cell channel 1: charger, discharger, protection, measurement



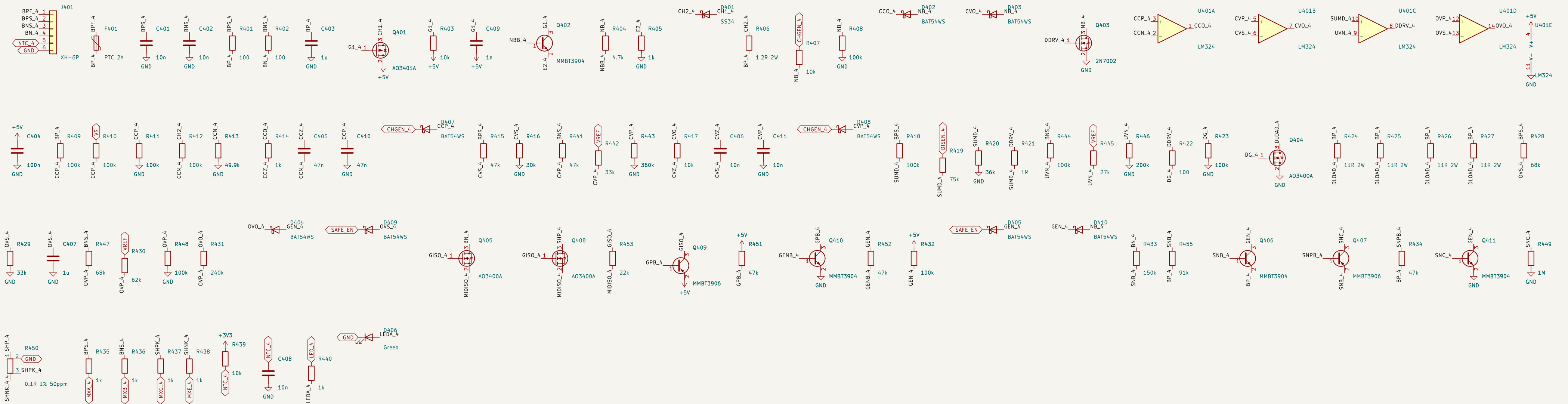
Cell channel 2: charger, discharger, protection, measurement



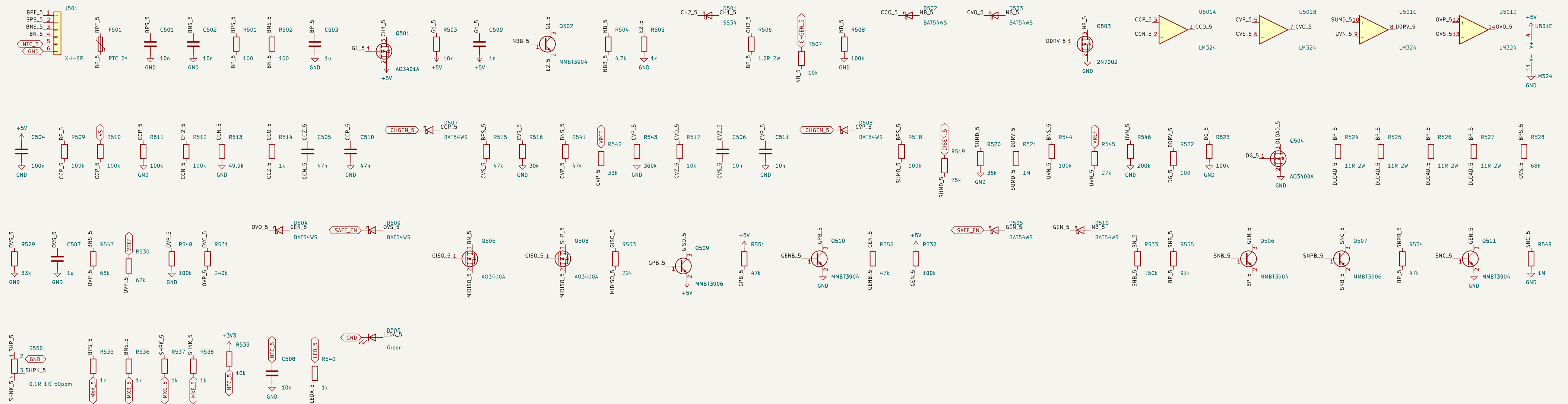
Cell channel 3: charger, discharger, protection, measurement



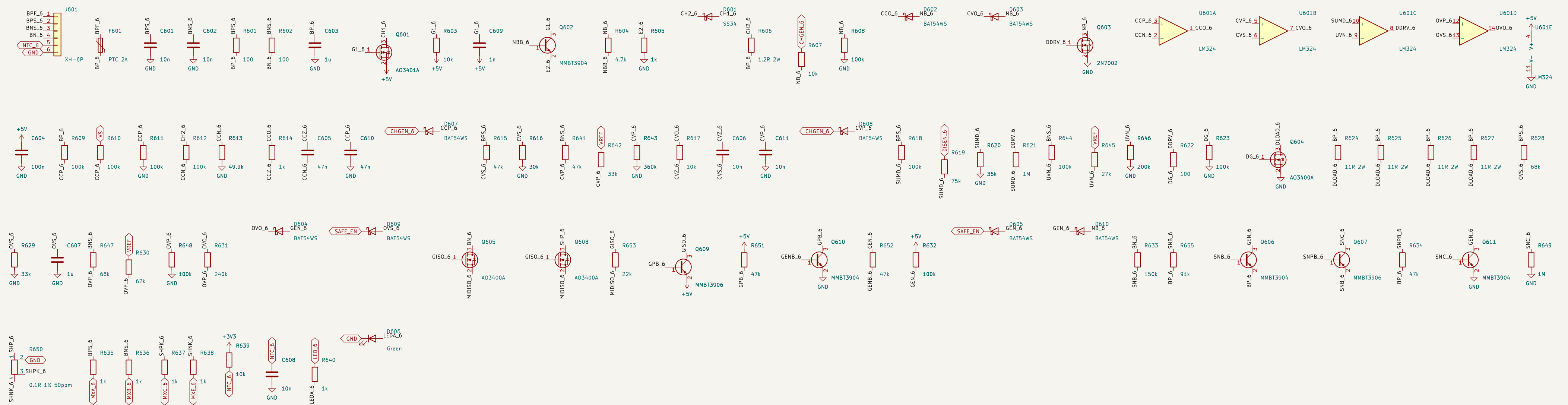
Cell channel 4: charger, discharger, protection, measurement



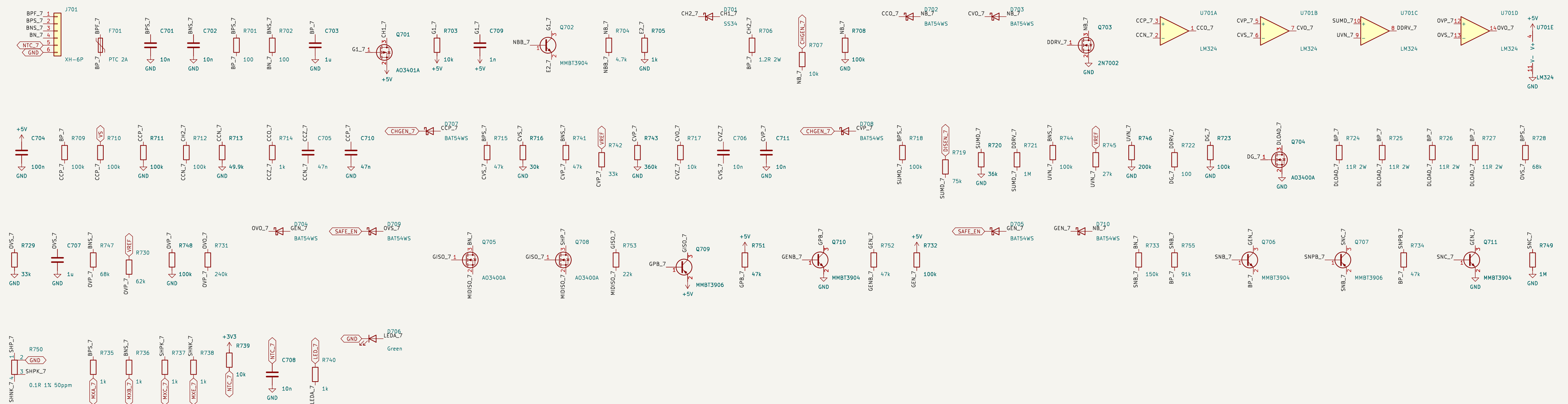
Cell channel 5: charger, discharger, protection, measurement



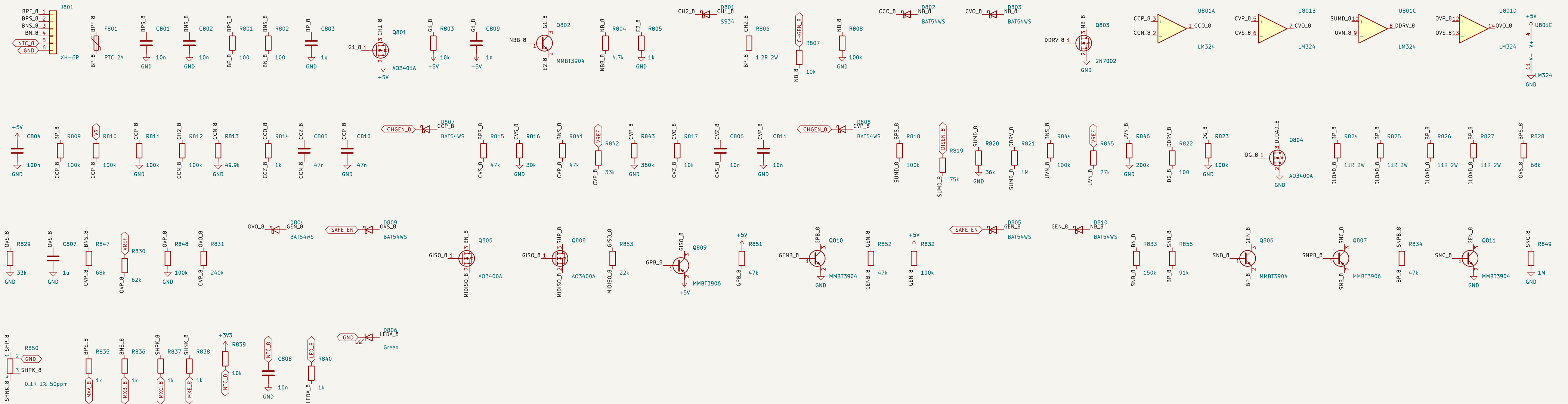
Cell channel 6: charger, discharger, protection, measurement



Cell channel 7: charger, discharger, protection, measurement



Cell channel 8: charger, discharger, protection, measurement



Mounting holes



Generated by hardware/gen/make_sch.py from the netlist DSL – do not edit by hand
Mounting holes
github.com/maxakash/lfppcb

Sheet: /mech/
File: lfp8_mech.kicad_sch

Title: LFP-8 8-channel LFP cell tester – mech

Size: A4	Date: 2026-10-01	Rev: 1.0
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 15/15